

What this project computed, and how far it can be trusted

A short account of the work behind *Extremality shift of rotating black holes at finite Gauss–Bonnet coupling*: the question, the quantity that answers it, the method, the result, and — at the same length — what the evidence does and does not support.

The question

A rotating black hole in Einstein gravity cannot carry arbitrary angular momentum at a given mass. For Kerr the bound is $M^2 \geq J$; in five dimensions, for the Myers–Perry solution with two equal angular momenta $J_1 = J_2 = J$, it is $M \geq \frac{3}{2}\pi^{1/3}J^{2/3}$ (arXiv:0801.3471). The solution that saturates it is extremal: its temperature vanishes and its horizon degenerates.

That bound belongs to the field equations, not to black holes in general. Add a higher-curvature term to the action and it moves. The question here is quantitative and narrow: *at fixed angular momentum, how does the extremal mass change when the coefficient α of a Gauss–Bonnet term is increased, beyond an infinitesimal perturbation of Einstein gravity?* The theory is

$$I = \frac{1}{16\pi G} \int d^5x \sqrt{-g} (R + \alpha \mathcal{L}_{\text{GB}}), \quad \mathcal{L}_{\text{GB}} = R^2 - 4R_{\mu\nu}R^{\mu\nu} + R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}, \quad (1)$$

the unique quadratic invariant whose field equations stay second order, and the first nontrivial Lovelock correction in five dimensions. Varying α means comparing equilibrium solutions of theories with different couplings, not a coupling that changes in time inside one spacetime.

Two lines of work make the sign of the displacement interesting. The weak gravity conjecture reads a shift of an extremality bound as a statement about what a black hole may decay into, and the Goon–Penco relation (arXiv:1909.05254) ties the same shift to the entropy correction of nearby non-extremal states. Both are normally stated for *charged* black holes, where $M \geq M_{\text{ext}}(Q)$ compares a mass with a gauge charge. The family studied here is uncharged and carries angular momentum instead; J is not a gauge charge and none of the spectrum arguments transfers. The Goon–Penco relation is therefore used only for what it is here, an identity of the extended first law, and no weak gravity consequence is drawn from it.

What was already known

There is no closed-form rotating black hole in Einstein–Gauss–Bonnet gravity, and the literature reaches finite coupling by two routes.

Perturbatively in α . Reall and Santos (arXiv:1901.11535) showed that the $\mathcal{O}(\alpha)$ thermodynamics follows without ever solving for the corrected metric; this was done for Myers–Perry in a general quadratic-curvature theory in arXiv:2405.04576, and the corrected metric in arXiv:2512.23797. For the equal-spin family, Ma, Li and Lü (arXiv:2009.00015) obtained the extremal mass at first order,

$$M_{\text{ext}} = \frac{3}{2}\pi^{1/3}J^{2/3} + \pi\alpha + \mathcal{O}(\alpha^2). \quad (2)$$

The initial slope is positive and equal to π . Equation (2) says nothing about the slope once α is no longer infinitesimal, and the slope can change a great deal while keeping its sign.

Numerically. Imposing $J_1 = J_2$ enhances the symmetry enough to factor out the angular dependence: the field equations become ordinary differential equations in the radius. Brihaye, Kleihaus, Kunz and Radu (arXiv:1010.0860) used that reduction to build the non-extremal equal-spin family at finite coupling, and obtained the entropy of its extremal limit algebraically from the near-horizon geometry. Kleihaus, Kunz and Radu (arXiv:2303.12471) then constructed the extremal solutions *directly*, in a gauge that puts the degenerate horizon at the origin, and mapped the domain of existence: the Einstein bound $j \leq 1$ survives at finite coupling, the static mass bound $M > 3\pi\alpha/4$ holds for the rotating solutions too, and the extremal set is conjectured to end at the nakedly singular static configuration with $M = 3\pi\alpha/4$, $J = S = 0$.

The gap, stated carefully. The honest version is narrower than the tempting one. Scale invariance makes the extremal branch a single curve in the reduced variables of arXiv:2303.12471, and that curve already fixes the extremal mass: with $\mu = M_{\text{ext}}/J^{2/3}$, $y = \alpha/J^{2/3}$, and x, j the published scaled coupling and spin, the definitions alone give

$$\mu = \mu(0)j^{-2/3}, \quad y = \frac{4x\mu}{3\pi}, \quad \mu(0) = \frac{3}{2}\pi^{1/3}. \quad (3)$$

So $M_{\text{ext}}(J, \alpha)$ is contained in the published domain of existence; what is not is its *derivative*. That function is not reported, and a slope read off a plotted domain boundary is not a quantity one would set beside the coefficient π of (2) — nor is differencing the masses of separately constructed extremal solutions, since each carries its own construction error. The derivative is what this project measures, and it is measured rather than differenced.

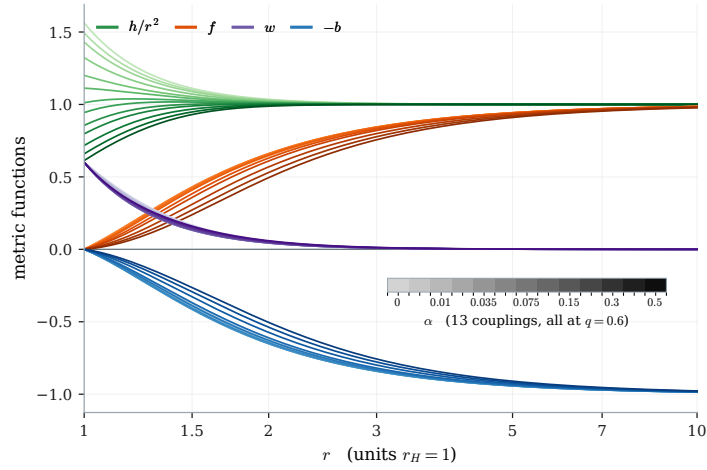


Figure 1: The four amplitudes of the equal-spin ansatz at all 13 couplings, $\alpha = 0$ (lightest) to $\alpha = 1/2$ (darkest), at spin $q = 0.6$ and $r_H = 1$; each curve is one nonlinear solve. Conventions follow fig. 1 of [arXiv:1010.0860](https://arxiv.org/abs/1010.0860): h is drawn as h/r^2 , which tends to unity at infinity, and b as $-b$, since b and f otherwise overlap.

The quantity that answers it

Promoting α to a thermodynamic variable gives the extended first law of Lovelock gravity, with a conjugate potential Ψ :

$$dM = T dS + 2\Omega_H dJ + \Psi d\alpha, \quad \boxed{\Psi = \left(\frac{\partial M}{\partial \alpha} \right)_{S,J}}. \quad (4)$$

The factor of two counts the two equal spins. “Potential” here is not a field: Ψ is one number per solution, measuring how its mass responds to a change of theory at fixed entropy and angular momenta.

Two facts make Ψ the right object. First, on a regular extremal branch the chain rule and (4) give $\partial M_{\text{ext}}/\partial \alpha|_J = T \partial S_{\text{ext}}/\partial \alpha|_J + \Psi$, and the first term drops as $T \rightarrow 0$ provided the extremal entropy derivative stays finite, so

$$\left. \frac{\partial M_{\text{ext}}}{\partial \alpha} \right|_J = \lim_{T \rightarrow 0} \Psi. \quad (5)$$

The slope is therefore obtained *pointwise* at each coupling, from one solution, and not reconstructed from a difference of masses. Second, the solver does not hold S and J fixed — it holds the horizon radius r_H and angular velocity Ω_H fixed — but the first law converts between the two,

$$\Psi = \left. \frac{\partial M}{\partial \alpha} \right|_{r_H, \Omega_H} - T \left. \frac{\partial S}{\partial \alpha} \right|_{r_H, \Omega_H} - 2\Omega_H \left. \frac{\partial J}{\partial \alpha} \right|_{r_H, \Omega_H}, \quad (6)$$

which subtracts from the total mass change the parts owed to entropy and spin and leaves the part owed to the change of theory.

How it was computed

The solutions. The equal-spin ansatz has four unknown radial amplitudes b, f, h, w . They are represented at Chebyshev–Lobatto nodes in a compactified radius, with the boundary behaviour factored out, and the discretised system is solved by damped Newton iteration at fixed (r_H, Ω_H, α) . The production dataset is 356 accepted solutions at 13 couplings from $\alpha = 0$ to $\alpha = 1/2$ in units $r_H = 1$; every one is a solve of the nonlinear problem at that coupling, not a perturbation about $\alpha = 0$. Figure 1 shows the four amplitudes at all 13 couplings at one spin. The methods are the standard ones for stationary black holes in higher dimensions, reviewed in [arXiv:1510.02804](https://arxiv.org/abs/1510.02804).

The response, without a step size. The three derivatives in (6) could be obtained by re-solving at $\alpha \pm \Delta\alpha$ and differencing, which trades a truncation error $\sim \Delta\alpha^2$ against a cancellation error $\sim \text{tolerance}/\Delta\alpha$, worst exactly where the solutions are hardest to converge. Instead, writing the discretised problem as $R_N(u; \alpha) = 0$ and using that it vanishes *identically* along the branch of roots,

$$A \frac{du}{d\alpha} = - \frac{\partial R_N}{\partial \alpha}, \quad A = \frac{\partial R_N}{\partial u}, \quad (7)$$

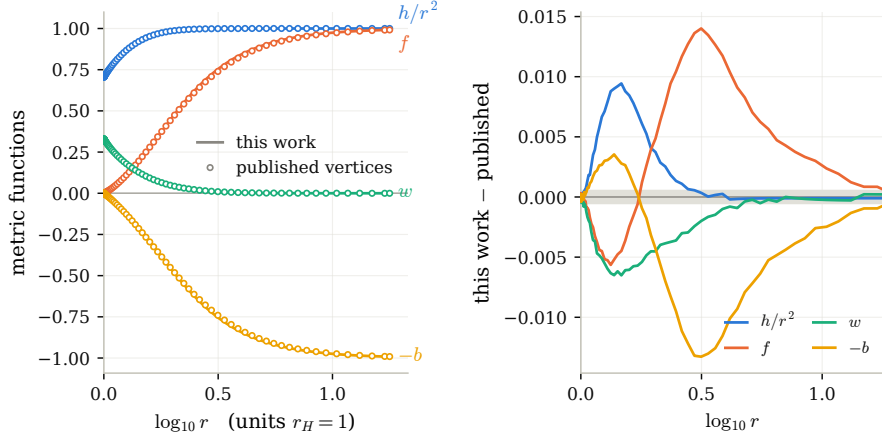


Figure 2: Left: the four metric functions at $\alpha_{\text{ref.}} = 3$, $q = 0.33$, ours as lines and as markers the 74 vertices read from fig. 1 of Brihaye, Kleihaus, Kunz and Radu, [arXiv:1010.0860](https://arxiv.org/abs/1010.0860). Right: the difference, with the band of one device unit of that figure (5.17×10^{-4}) shaded. Conventions as in fig. 1, which is what makes the two directly comparable.

which is the implicit function theorem. One linear solve with the Jacobian the last Newton step already used returns $du/d\alpha$ exactly, with no step to refine; propagating it through the charge functionals gives Ψ . Local partial derivatives are taken by complex step. Against real central differences the response agrees at the second-order rate, the error ratios along a halving ladder falling between 3.951 and 4.028.

The extremal limit. Following each family towards $T = 0$, the coldest state reached has scaled temperature $\tau = TJ^{1/3}$ down to 3.34×10^{-6} , and the quoted extremal values are $T = 0$ intercepts of cubic fits through the six coldest states. *No solution is constructed at $T = 0$:* every extremal number is an extrapolant, and the distance each fit had to travel is reported beside it. The quoted bands are the spread of the intercept over twenty-one fitting variations (degrees one to three, six- and twelve-point windows, the coordinates T and τ , two non-polynomial forms): sensitivities to stated modelling choices, not confidence intervals, and blind to an error common to all.

Reproducing the published family

Before measuring anything new, the solver has to return the family [arXiv:1010.0860](https://arxiv.org/abs/1010.0860) already constructed. Figure 2 is that comparison, at their coupling $\alpha_{\text{ref.}} = 3$ — which is $\alpha = 0.75$ here, the conversion $\alpha_{\text{ref.}} = 4\alpha$ being itself read off their action rather than assumed — and spin $q = 0.33$.

The published figure cannot be redistributed — its licence covers arXiv and not third parties — so neither it nor the source file is in this repository; the coordinates are. The arXiv source ships it as gnuplot PostScript, every curve a polyline carrying its own vertices, so nothing is digitised: the device-to-data map is fitted from the file’s own printed ticks, refused unless those are collinear to within the one device unit gnuplot rounds to, and the four curves are read off, 74 vertices each.

At the horizon the two calculations agree to 2.07×10^{-4} , a third of one device unit, which fixes the parameters and the boundary conditions. In the interior they part by up to 1.40×10^{-2} , a smooth excursion peaking where each curve is steepest. That departure is not ours: the same figure draws a near-Einstein coupling at which our solution is Myers–Perry to thirteen digits, and there its published curve sits 8.19×10^{-3} from ours and 9.54×10^{-3} from the *closed form*, which carries no numerical error at all. The cause is not established here and none is asserted; the comparisons this work rests on, listed next, close at the digits quoted there.

Four comparisons with the same paper are quantitative rather than pictorial and carry the digits a figure cannot. The ansatz is identically their eq. (2.8) in the gauge $g_{\theta\theta} = r^2$; the charge normalisations are those of eq. (3.6), with J the momentum in each plane; their entropy, eq. (3.10), equals our Jacobson–Myers form exactly when $\alpha_{\text{ref.}} = 4\alpha$, which is how that conversion is fixed; and our rederived near-horizon solution satisfies eqs. (4.11) and (4.12) as printed, to 9.17×10^{-14} and 6.66×10^{-16} , its extremal entropy matching the $T \rightarrow 0$ limit of our bulk solutions to 4.06×10^{-6} . One comparison does not close, and is reported below rather than absorbed.

What came out

The slope $\partial M_{\text{ext}}/\partial\alpha|_J$ is positive across the whole sampled range, so the extremal mass at fixed angular momentum increases with the coupling and the bound on the scaled spin tightens. It is not constant: it falls

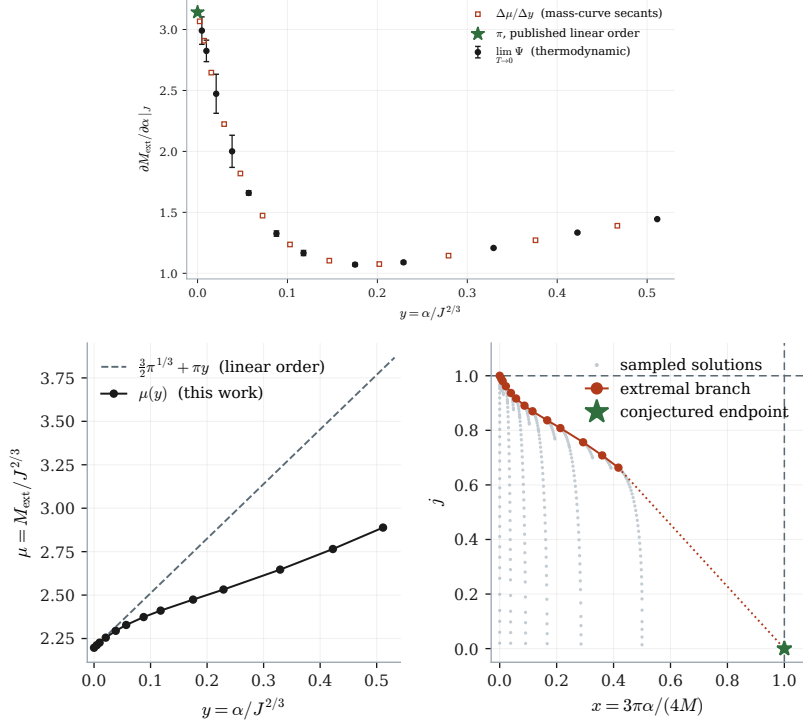


Figure 3: Top: the measured slope against $y = \alpha/J^{2/3}$, with the published linear-order value π as a star and secants of the mass curve as open squares. Bottom left: the extremal mass the slope integrates to, against the first-order line. Bottom right: the same branch in the (x, j) domain-of-existence plane of [arXiv:2303.12471](#). Circles are $T = 0$ intercepts of fits, not constructed solutions.

well below the perturbative value π , reaching 1.07 near $y = 0.175$, and then rises again to 1.44 at $y = 0.511$. Both moves exceed the fit sensitivities of the rows involved by more than an order of magnitude. Where the turning point lies *between* the sampled couplings is not resolved, which is why we quote a lowest sampled value and not a minimum.

Integrating the slope gives the mass itself: over the sampled range μ rises by 31%, while (2) runs above it and overshoots by a factor 1.32 at the largest coupling. The fall of the slope is a fall in the *rate* — the extremal mass grows more slowly than linear order predicts, not more slowly than at $\alpha = 0$.

Three ways to read it. The result becomes less abstract when restated.

1. *The spin bound, and the horizon.* By (3) a positive μ' is the same statement as a decreasing j_{ext} , so the bound $j \leq 1$ reported in [arXiv:2303.12471](#) is the integrated form of the sign measured here pointwise. Equivalently, setting $T = 0$ in the Smarr relation gives $\alpha\Psi_{\text{ext}} = M_{\text{ext}} - 3\Omega_H J$, so a positive slope is exactly $M_{\text{ext}} > 3\Omega_H J$: the two sides coincide at $\alpha = 0$, first order says the excess opens linearly, and the measurement says it keeps opening at a rate that falls and then recovers.
2. *The geometry.* At fixed T and Ω_i the Gibbs potential is the on-shell action per unit time, whose only explicit α -dependence is the Gauss–Bonnet term, so $\Psi = -(16\pi)^{-1} \int_{\Sigma} \sqrt{-g} \mathcal{L}_{\text{GB}} d^4x$ over a stationary exterior slice: a positive Ψ is a negative integrated Gauss–Bonnet density. With $\partial S / \partial \alpha|_{M,J} = -\Psi/T$ it is likewise a decrease of entropy at fixed mass and spin — compatible with entropy per spin along the *extremal* branch growing by 4.8, because extremal states at fixed J have different masses as α changes.
3. *The turning point.* With $\omega = \Omega_H J^{1/3}$, the same Smarr relation gives $3\omega = \mu - y\mu'$ and hence $3\omega' = -y\mu''$. For $y > 0$ the inflection of the extremal mass is exactly the stationary point of ω , and the minimum of the slope is a *maximum* of ω : the extremal horizon with the largest angular velocity in units of $J^{-1/3}$. That coupling is not otherwise distinguished, and is not a boundary of the domain of existence.

Why the slope has to rise. The bound $M > 3\pi\alpha/4$ is $\mu(y) > 3\pi y/4$. If μ' stayed below some $c < 3\pi/4$ everywhere, then $\mu(y) \leq \mu(0) + cy$, which violates the bound once $y > \mu(0)/(3\pi/4 - c)$. So μ' must return to at least $3\pi/4 = 2.356$, above every slope measured here: the observed rise is required, not accidental. If the branch also ends as conjectured in [arXiv:2303.12471](#), the average slope tends to that value. Our range stops well short — the largest extremal x is 0.4170, against a conjectured endpoint at $x = 1$ — so this is consistency, not evidence.

How far it can be trusted

The response solve has no analytic counterpart, so it is checked against four things that do not use it.

check	what it tests	worst deviation
Boulware–Deser static limit	M, T, S, Ψ in closed form at finite α	2.60×10^{-9}
Published perturbative $\Psi_0(q)$	the response at $\alpha = 0$, at every spin	2.86×10^{-8}
Smarr relation as an equation for Ψ	charges only: no Jacobian, no linear solve	1.45×10^{-6}
On-shell Gauss–Bonnet action	curvature only: not even the charge extraction	4.67×10^{-7}
Near-horizon entropy function	S_{ext}/J at finite coupling, algebraically	4.06×10^{-6}
First law in the spin direction	internal consistency of the observables	1.35×10^{-8}
$G_{\mu\nu} + \alpha H_{\mu\nu} = 0$ off-grid	equation closure, independent evaluator	9.76×10^{-9}
Einstein extremal limits	$\mu(0)$ and $\Psi_{\text{ext}}(0)$ against $\frac{3}{2}\pi^{1/3}$, π	4.56×10^{-11} , 1.19×10^{-7}
Radial resolution, 26 full re-walks	the whole pipeline at two coarser N	6.89×10^{-5}

The last two rows deserve comment. The Einstein limit is recovered from the same continuation and extrapolation used at finite coupling, with the analytic constants imposed nowhere, so agreement there does not establish the error at finite y . And the resolution study, which re-ran the complete ascending sequence and $T = 0$ extrapolation at two coarser resolutions at every coupling, moves the intercept by less than the fit sensitivity in all 26 sequences: that bounds the resolution effect without proving convergence.

Limitations. The extrapolation is the leading one, and the bands around it are sensitivities, not error bars. The coupling coverage is finite, the spin coverage uneven, and the work is restricted to equal spins and to the branch continuously connected to Myers–Perry; nothing here shows that branch is unique, stable, or the lowest in mass at the same angular momenta. No check closes the gap that matters most: the mass-curve comparison shares systematics with the potential, the near-horizon route gives entropy rather than mass, and the Smarr and on-shell routes remove the response solve but not the solutions. Finally, this is classical Einstein–Gauss–Bonnet with the action (1): in an effective field theory, terms beyond Gauss–Bonnet enter at the same order as the higher powers of α kept here, so the non-monotonic slope is not a universal prediction of a derivative expansion.

One comparison that does not close. A separate reconstruction of the non-extremal family gives an outer ergosurface radius 1.102101 against the 1.104 quoted in [arXiv:1010.0860](#), at $\alpha = 1/2$ in our convention. It is unresolved, recorded as a declared negative result rather than absorbed into a tolerance, and not offered as an erratum. It tests the metric functions away from the horizon and touches neither the charge extraction, the response solve nor the extrapolation.

New, reproduced, and cited

New here: Ψ measured on rotating solutions at finite coupling across the whole family, with the sign change it undergoes; the extremal slope obtained pointwise at 13 couplings, its minimum and rise resolved; the extremal mass curve it integrates to and its departure from (2); the reading of Ψ as an on-shell Gauss–Bonnet action, tested against the response on 43 archived solutions; and the identification of the slope minimum with the maximum of $\mathcal{O}_H J^{1/3}$.

Reproduced, not new: the equal-spin family of [arXiv:1010.0860](#) and its profiles; the closed-form static solution; the $\mathcal{O}(\alpha)$ potential implied by [arXiv:2405.04576](#) and the first-order extremal mass π of [arXiv:2009.00015](#); the near-horizon extremal entropy of [arXiv:1010.0860](#); and the extremal branch itself, which [arXiv:2303.12471](#) constructs directly and this work reaches by extrapolation, with its bounds $j \leq 1$ and $x < 1$.

Cited, not reproduced: the Myers–Perry and Boulware–Deser solutions (oracles, checked against the solver but not rederived), Lovelock’s theorem, the Wald and Jacobson–Myers entropy formulae, Sen’s formalism, the covariant phase space derivation of the extended α -term, and the Goon–Penco universality claim, of which only the first-law identity is used, and rederived.

The apparatus. One discipline shaped the repository more than any other: **no number that appears in the paper is typed into the paper**. Every figure, table and quoted quantity is generated from the recorded measurement into `numbers.tex` and read back as a macro, so nothing can drift; this document draws on the same file. Around it sit a static checker on macros, literal decimals and citations; a provenance record per result, naming its command and its hashed inputs and audited against the bytes on disk; and a declared list of failing checks held equal by a test to the actual ones, so it cannot become a way of muting the audit. Three findings of the process are worth recording, in each of which the obvious diagnostic measures something other than it appears to: a different linear-algebra stack moved an extrapolated intercept by more than either genuine resolution change, the walk being adaptive; the on-shell quadrature diverges past about eighty nodes, $\mathcal{L}_{\text{GB}}$ being a cancellation that decays like r^{-8} while the compactification amplifies what survives; and a comparison is only as sharp as the weaker side’s resolution.